IEORE4004: Optimization Models and Methods - Prof. Yuri Faenza 11/18/2025

Assignment 5

Deadline: 12/02/2025, h: 11:59pm

- This assignment sheet has 3 exercises. You must submit your solution via Gradescope. See the syllabus for the late submission policy.
- You are allowed to discuss the assignment with others and to use AI tool but the write-up must be individual work. Please mention in your write-up all the people you have discussed the solution with, and how you used AI (if you did).

1) Greedy Algorithms

Problem 1: Recall that an *independent set* of a graph $G(V, E)$ is a set $S \subseteq V$ such that no two nodes of S are *adjacent*[1] in G. The *Maximum Independent Set problem* (MIS) is defined as follows:

a) Given: A graph $G(V, E)$;

b) Find: An independent set of G of maximum cardinality.

Consider the following greedy algorithm for the problem of finding a MIS on a graph $G(V, E)$:

1. Let $S = \emptyset$.
2. While G has still a node:
   (a) Find a node v of G that is adjacent to the smallest number of nodes of G.
   (b) Add v to S.
   (c) Remove v and the nodes adjacent to v from G.
3. Output S.

a) Argue that the algorithm correctly produces an independent set.

b) Give an example where the algorithm does not give an optimal solution.

b) Dynamic Programming

Problem 2: Consider the modification of the Investment problem seen in class and in lecture notes #13, Section 5. We can invest \$6 and each investment must be a multiple of \$1. There are three possible investments:
Investment 1: investing $x_1 > 0$ dollars yields $c_1(x_1) = 7x_1 + 2$ dollars, where $c_1(0) = 0$.
Investment 2: investing $x_2 > 0$ dollars yields $c_2(x_2) = 3x_2 + 7$ dollars, where $c_2(0) = 0$.
Investment 3: investing $x_3 > 0$ dollars yields $c_3(x_3) = x_3^2 - 5x_3 + 5$ dollars, where $c_3(0) = 0$.

For each of the questions below, explicitly report the recursive function, the table, and the optimal solution.

a) Write a recursive function for the dynamic program that solves the problem above. Compute the value of the optimal solution.

b) Compute the optimal solution to the problem in a), i.e., how much money we should invest in each investment.

c) Consider the modification of the problem in a) where we are only allowed to invest up to 2\$ in Investment 3, and compute the optimal solution and its value.

[1] Two nodes are adjacent if they are connected by an edge.

Problem 3:

a) You are given a 4×7 board and a robot initially placed at the upper left corner. On each step, the robot can either move one cell to the right (R), or one cell down (D). We are interested in the number of ways there are for the robot to move to the destination, which is the lower right corner.

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For example, two possible routes the robot can take are:

$$\begin{aligned} Route1 &: R, R, R, R, R, R, D, D, D \\ Route2 &: R, D, R, D, R, D, R, R, R \end{aligned}$$

Use Dynamic Programming to calculate how many distinct ways there are for the robot to move to the destination. (Hint: You can define rows as stages and columns as states; and $f(i, j)$ represents the number of ways there are if the board is of size $i \times j$.). Explicitly report the recursive function, the table, and the optimal solution.

b) Now assume that coins are placed in some cells (shown in the table below). The goal of the robot is to collect as many coins as possible and bring them to the destination. As in the previous part, the robot can only move one cell right or one cell down each time. Use Dynamic Programming to figure out which route the robot should take to maximize the amount of coins collected.

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Explicitly report the recursive function, the table, and the optimal solution.